DEMO-READY WORKFLOW - MICROSOFT FOUNDRY GPT-SOL EDITION

BidSight Run and Demo Guide

Step-by-step local setup, Foundry configuration check, sample procurement walkthrough, and presentation script.

Audience	Developer, evaluator, interviewer, or demo presenter
Demo duration	8 minutes plus setup
Sample folder	simple-data/
Updated	11 August 2026

Decision-support principle

Microsoft Foundry gpt-sol interprets quotation language and explains verified results. Deterministic Python code performs compliance checks, arithmetic, scoring, and ranking. The final purchasing decision remains with an authorised user.

Prepared for the BidSight portfolio project by Badar Butt

1. What the audience should understand

- Foundry gpt-sol reads and structures quotation content; it does not calculate procurement scores.
- A human reviews every extracted value before evaluation.
- Python enforces mandatory requirements and calculates reproducible scores.
- Non-compliant vendors remain visible so rejection reasons stay transparent.
- The final recommendation is advisory; the purchasing decision remains with the user.

1.1 Demo flow

1. Define	2. Upload	3. Review	4. Score	5. Recommend
Baseline and requirements	Three vendor PDFs	Editable Foundry fields	Python rules	Foundry explanation

1.2 Roman Urdu summary

Short explanation

Pehle requirements set hongi, phir teen quotations upload hongi. Foundry gpt-sol data extract karega, user values confirm karega, Python scores calculate karega, aur akhir mein Foundry verified results ko explain karega.

2. One-time local setup

2.1 Backend dependencies

```
cd backend
uv sync --group dev
```

The provider dependency is now the OpenAI Python SDK. uv sync will refresh backend/uv.lock.

2.2 Frontend dependencies

```
cd frontend
pnpm install
```

2.3 Backend environment file

Copy the root .env.example to a new untracked root .env and fill in only your real local Foundry values.

```
DATABASE_URL=sqlite:///./bidsight.db
FOUNDRY_ENDPOINT=https://YOUR-RESOURCE-NAME.services.ai.azure.com
FOUNDRY_API_KEY=your_real_key_here
FOUNDRY_MODEL_DEPLOYMENT=gpt-sol
FOUNDRY_REQUEST_TIMEOUT_SECONDS=90
UPLOAD_DIR=uploads
FRONTEND_URL=http://localhost:3000
MAX_UPLOAD_SIZE_MB=10
```

2.4 Frontend environment file

```
# frontend/.env.local
NEXT_PUBLIC_API_URL=http://localhost:8000
```

Security rule

Never paste the real Foundry key into README.md, .env.example, source code, a PDF, a screenshot, or a public repository. Only the local untracked .env should contain it.

3. Start and verify BidSight

3.1 Terminal 1 - FastAPI

```
cd backend
uv run uvicorn app.main:app --reload
```

Address	Expected result
http://localhost:8000	BidSight API status JSON
http://localhost:8000/api/health	{"status":"ok"}
http://localhost:8000/docs	Interactive Swagger API documentation

3.2 Terminal 2 - Next.js

```
cd frontend
pnpm dev
```

Open http://localhost:3000. Go to Settings and confirm the API badge reports Connected.

3.3 Optional pre-demo checks

```
cd backend
uv run pytest

cd ../frontend
pnpm typecheck
pnpm lint
pnpm build
```

Presenter tip

Start both servers before the audience joins. Keep the frontend, Swagger docs, and simple-data folder open. Confirm Foundry is configured without showing the key.

4. Create the sample evaluation

Select New Evaluation and enter these values.

Field	Demo value
Evaluation title	Computer Lab Laptop Procurement
Product/category	IT Equipment
Quantity	25
Maximum budget	PKR 4,000,000
Required delivery	14 days
Notes	Business laptops for a new teaching laboratory

4.1 Mandatory requirements

Requirement	Value	Unit	Rule
Processor	Intel Core i5 or equivalent	-	Mandatory match
Minimum RAM	16	GB	At least
Minimum storage	512	GB	At least
Minimum warranty	24	months	At least
Delivery	14	days	At most

Do not force a winner

Do not weaken requirements or edit source values to favour TechCore. The winner must emerge from the quotation evidence and deterministic eligibility rules.

5. Upload and review the quotations

Upload these three files from simple-data/ and do not upload the requirements PDF as a vendor quotation.

- techcore-solutions-quotation.pdf
- digital-systems-quotation.pdf
- future-computers-quotation.pdf

5.1 What happens after processing

- FastAPI validates PDF extension, content type, header, file size, and the three-file maximum.
- PyMuPDF extracts page-by-page text from the digitally generated PDF.
- Foundry gpt-sol returns a typed Pydantic response; missing fields remain null.
- The saved PDF remains available if Foundry processing fails, allowing Retry.

5.2 Review checklist

Area	Confirm
Supplier	Vendor name, product name, product model
Commercial	Quantity, unit price, subtotal, tax, total price, currency
Fulfilment	Delivery days, warranty months, payment terms, support
Technical	Processor, RAM, storage, operating system, other specifications

Human review gate

Compare the form with the source PDF and correct any extraction error. Leave unknown information blank; never invent a value. Confirm every quotation before scoring.

6. Expected result

Vendor	Total	Delivery	RAM	Warranty	Status
TechCore Solutions	PKR 3,625,000	10 days	16 GB	36 months	COMPLIANT
Digital Systems	PKR 3,450,000	18 days	16 GB	24 months	NON_COMPLIANT
Future Computers	PKR 3,300,000	7 days	8 GB	12 months	NON_COMPLIANT

6.1 Why TechCore should be recommended

- TechCore meets processor, RAM, storage, quantity, delivery, and warranty requirements.
- Digital Systems is cheaper but 18-day delivery exceeds the mandatory 14-day maximum.
- Future Computers is cheapest but 8 GB RAM and 12-month warranty fail mandatory minimums.
- Only eligible positive prices set the price-score baseline.

Expected recommendation

TechCore Solutions should emerge as the recommended compliant vendor. Foundry wording may vary, but it cannot change Python-generated status, scores, or rank.

6.2 What to show on comparison

- Recommended-row emphasis and visible failed vendors.
- Total price, compliance, delivery, warranty, price score, overall score, and status.
- Failure explanations and cheaper-vendor tradeoff.
- Strengths, risks, and missing information in the recommendation panel.

7. Suggested 8-minute presentation

Time	Screen	Talking point
0:00-0:45	Dashboard	Decision support for quotation evaluation; not a chatbot.
0:45-1:45	New Evaluation	Show baseline and mandatory versus preferred requirements.
1:45-2:45	Upload	Upload three PDFs; explain validation and the three-file limit.
2:45-4:30	Review	Show editable fields and emphasise human confirmation.
4:30-6:15	Comparison	Point out deterministic scores and mandatory failures.
6:15-7:30	Recommendation	Foundry receives verified evidence and existing scores only.
7:30-8:00	Close	Read the advisory disclaimer and state the MVP scope.

7.1 Strong closing statement

Demo close

BidSight combines AI-assisted document understanding with deterministic procurement controls. Every extracted value is reviewable, every failure remains visible, and purchasing authority stays with the user.

7.2 Questions to expect

- Why not let Foundry score? Python rules are reproducible, testable, and auditable.
- Why keep failed vendors? Reviewers need transparent comparison and failure evidence.
- What if a value is missing? It is unknown/missing, never assumed compliant.
- Which PDFs work? Digitally generated text-readable PDFs; OCR is future work.

8. Troubleshooting and final checklist

Symptom	Action
API unavailable	Start FastAPI and verify NEXT_PUBLIC_API_URL.
Settings not connected	Open /api/health and check FRONTEND_URL.
Foundry endpoint missing	Set FOUNDRY_ENDPOINT to the resource endpoint in root .env.
Foundry key missing	Set FOUNDRY_API_KEY only in the untracked root .env.
Deployment not found	Match FOUNDRY_MODEL_DEPLOYMENT exactly to gpt-sol in Foundry.
PDF rejected	Use a valid text-readable .pdf under 10 MB.
Scoring conflict	Process and confirm every quotation before comparison.
Recommendation unavailable	Confirm an eligible vendor and valid Foundry configuration.

8.1 Five-minute pre-demo checklist

- [ ] Backend root, /api/health, and /docs open successfully.
- [ ] Frontend opens and Settings shows Connected.
- [ ] Foundry endpoint, key, and gpt-sol deployment are configured without showing the key.
- [ ] The three quotation PDFs are ready in simple-data/.
- [ ] No incomplete evaluation will confuse the demonstration.
- [ ] Vendor facts match simple-data/expected-results.json.

8.2 Optional clean reset

Stop both servers. If you intentionally want a clean demo, remove only the local backend database and generated evaluation subfolders under backend/uploads; keep backend/uploads/.gitkeep. This reset is not recoverable unless backed up.

> **Disclaimer**
>
> BidSight provides procurement decision support. AI-generated recommendations are advisory, and final purchasing decisions remain with authorised users.